HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)

Design and Analysis of Knee Rehabilitation Robot

Preliminary Design

- Studied different existing systems in the same domain.
- Determined various parametes for the knee rehabilitation process with the help of sample data collected.
- Designed & compared various systems using existing mechanisms.

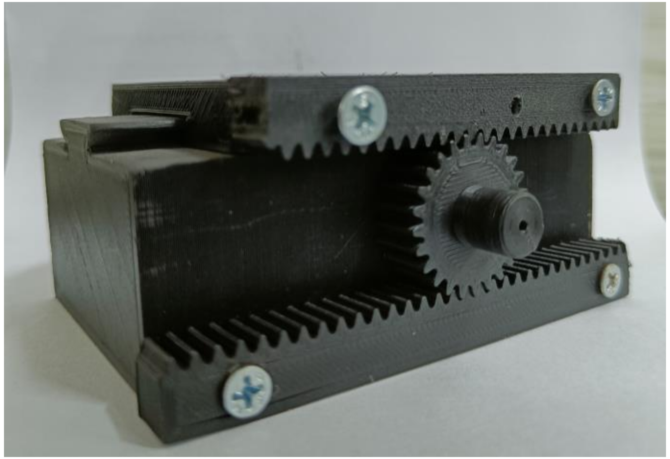


Figure1:3D Printed design

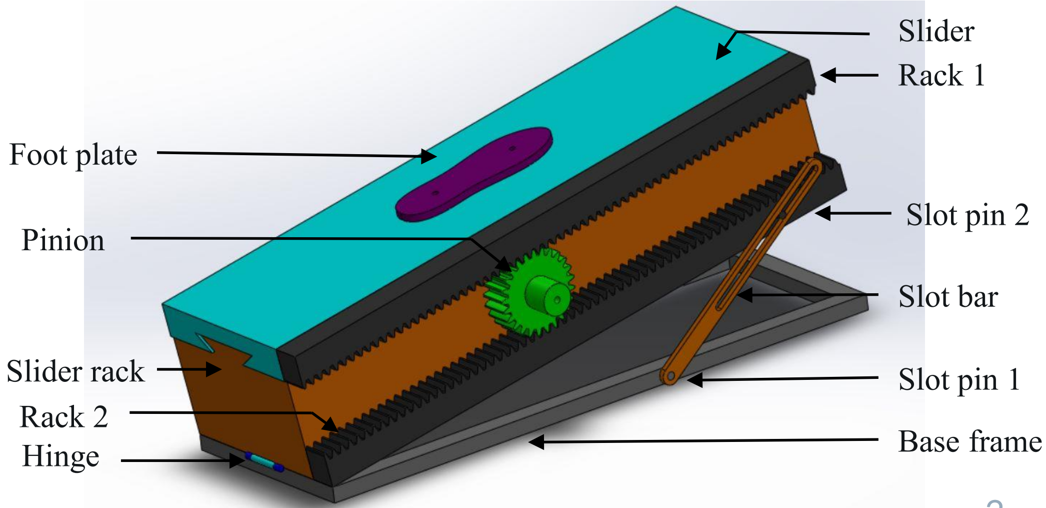


Figure2 : Solidworks design

Preliminary Actuation Design Study

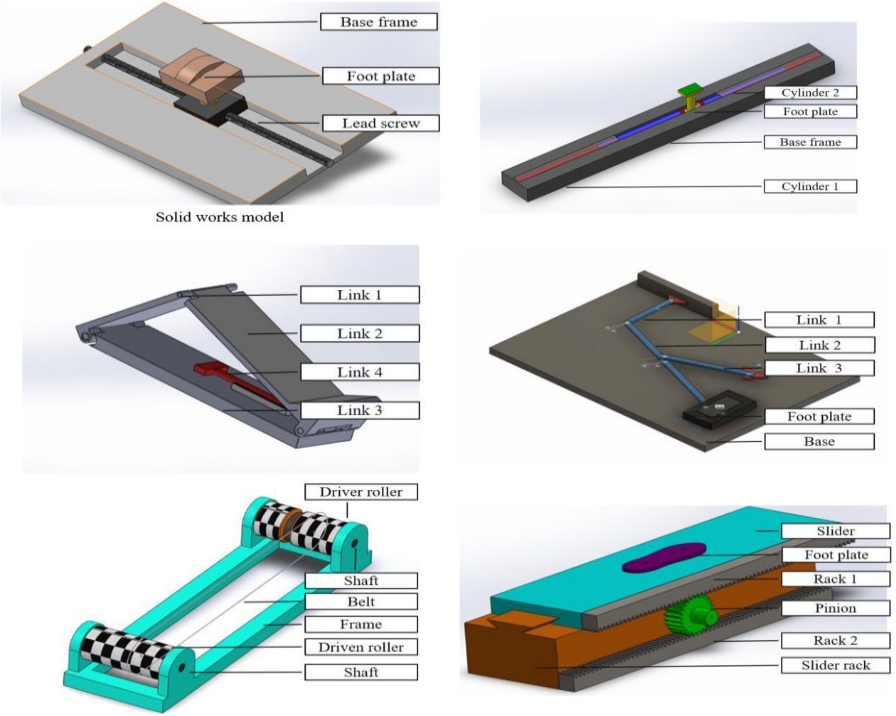


Figure3 : CAD of Linear mechanims

Factors	**Roller based linear actuation**	**Slider based linear actuation**	**Four bar mechanism based**	**Grasshopper mechanism based**	**Lead Screw mechanism based**	**Hydraulic Cylinder based**
Motion	Non linear	Purely linear	Purely linear	Linear	Purely linear	Purely linear
Power consumption	High	Low	Low	Low	Low	Very high
Slippage	Yes	No	Yes	Yes	No	No
Complexity	High	Low	High	High	Low	Moderate
Parts	Too many parts	Less parts	Too many parts	Too many parts	Too many parts	Many
Links	More	Less	More	More	More	More
Cost	High	Low	Moderate	Moderate	Moderate	Very high
Maintenance	High	Moderate	Low	Low	Moderate	Very high
Motion precision	Less accurate	Highly accurate	Moderately accurate	Moderately accurate	Highly accurate	Moderately accurate
Mechanism	Roller and belt	Rack and pinion	4- bar mechanism	Modified Scott-Russel's mechanism	Lead and screw	Hydraulic actuation

Table1 : Design comparrisons of Linear mechanims

1st Conceptual design

CAD model of Single rack and pinion with single slotted bar

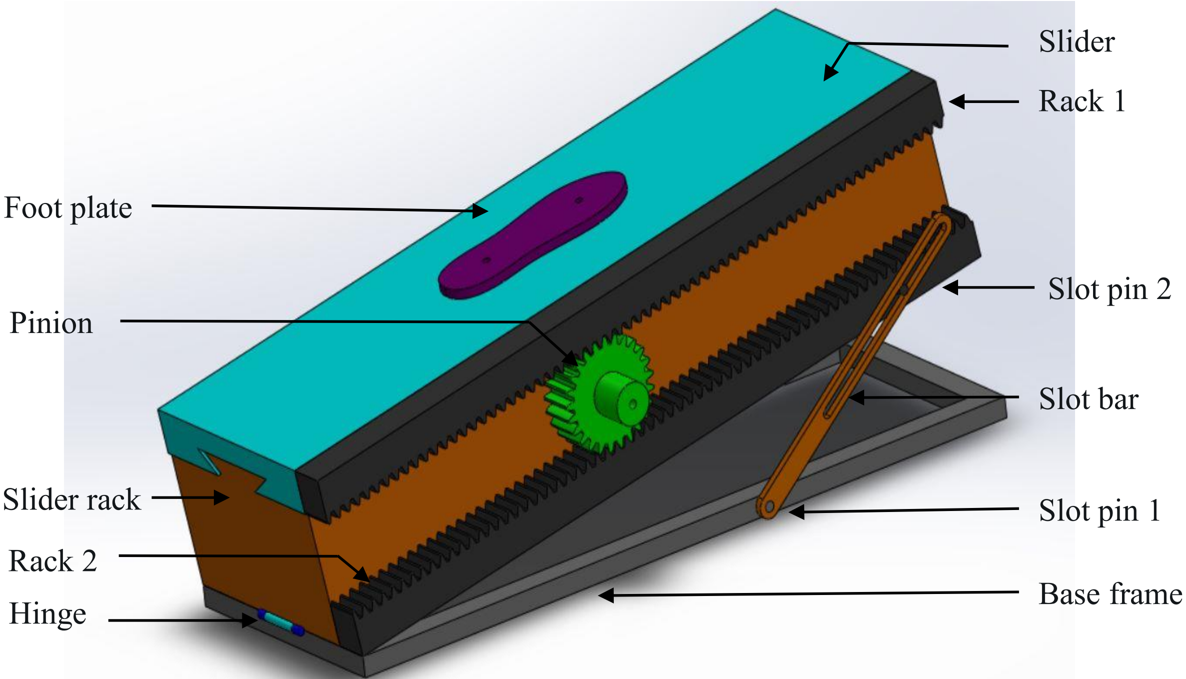


Figure4 : solidworks design

Structural analysis of 1st Conceptual design

RESULTS

Reasons of failure

- Unbalances load distribution
- Motion constrain because of misaligned slot bar

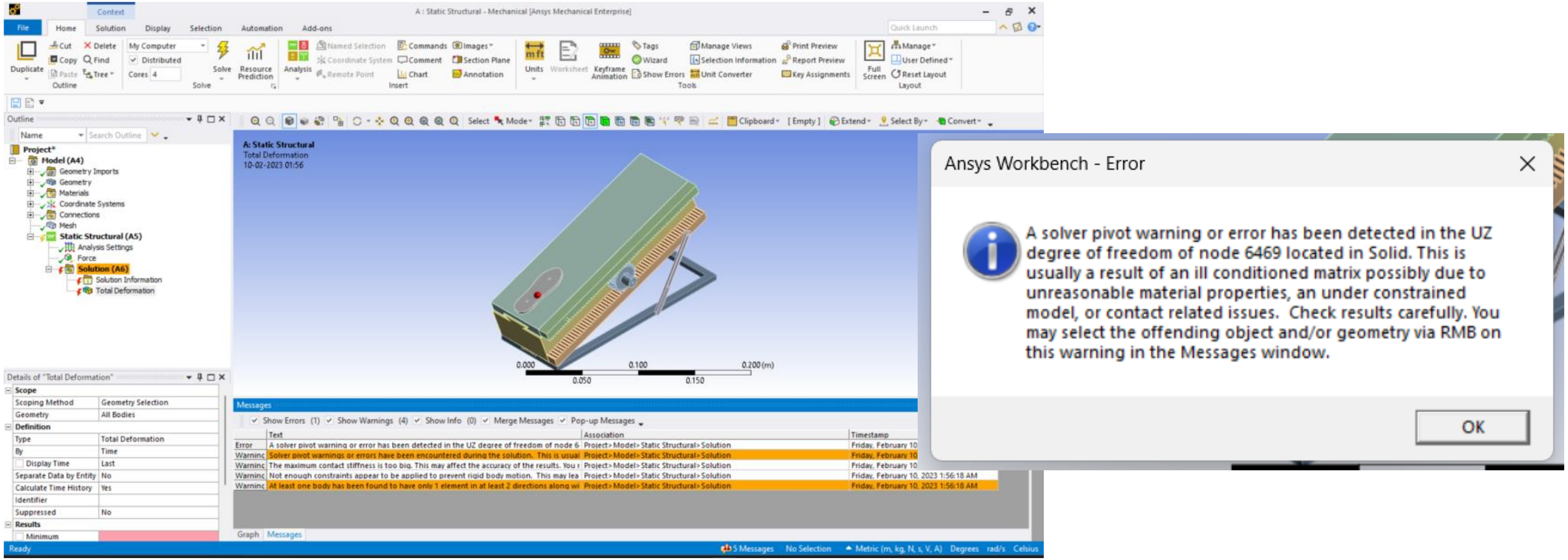


Figure5 : ANSYS interface of design 1

2^{nd} modified design

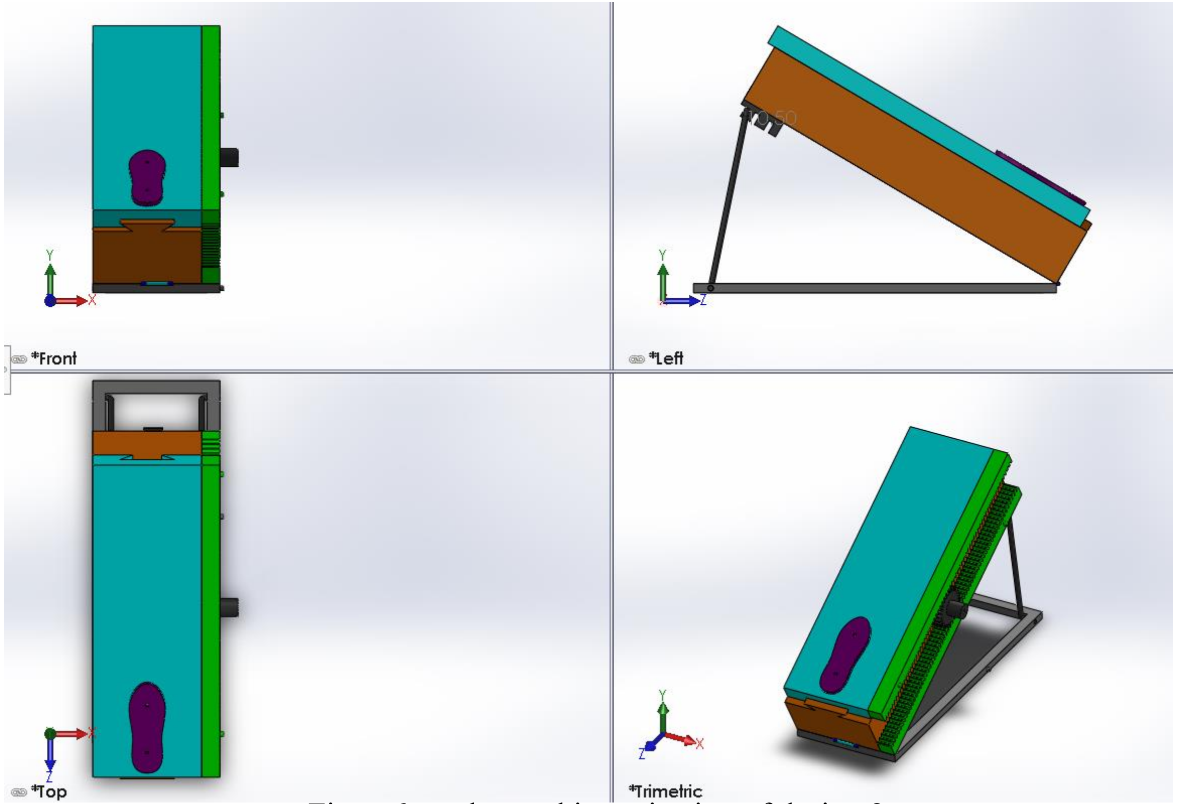


Figure6 : orthographic projection of design 2

Structural analysis of 2nd modified design

RESULTS

- ▷ Only 1dof is there in the above modified design.
- ▷ Unbalanced torque generated at one end of the rack.

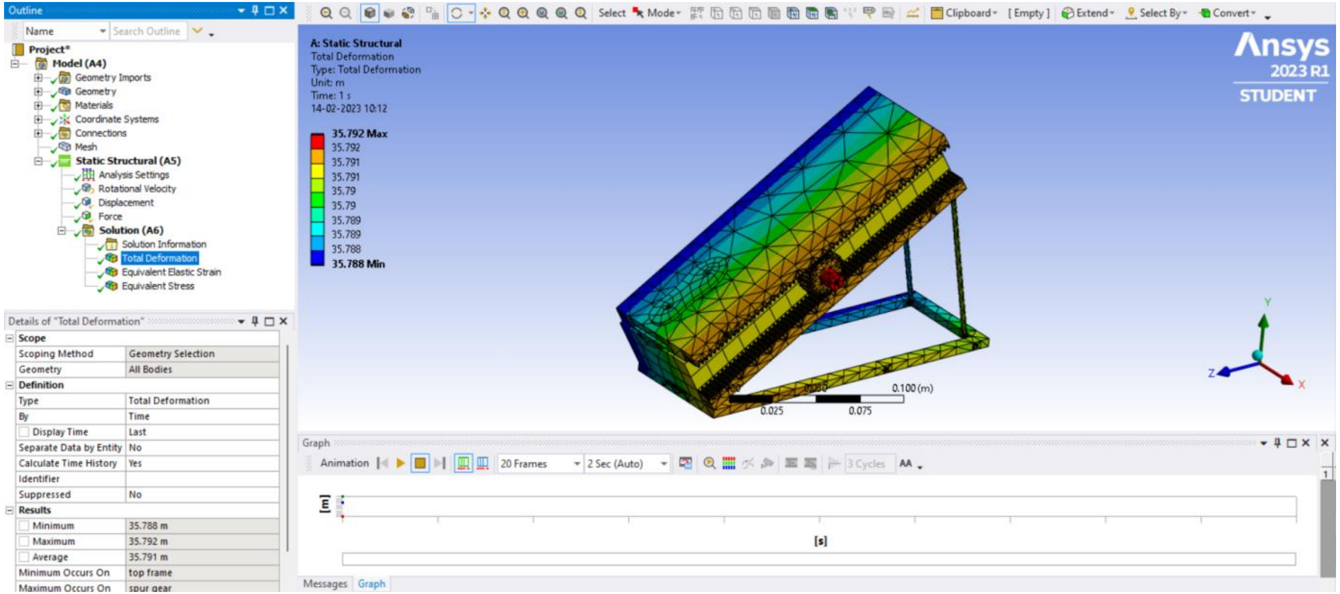


Figure7 : ANSYS interface of design 2

Linear Actuation - Slider rack mechanism

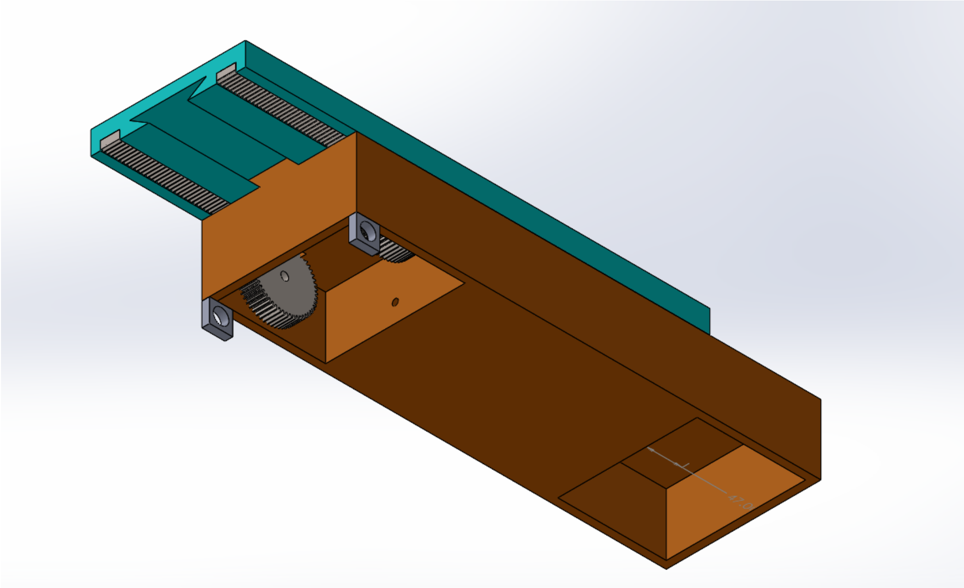


Figure8 : Isometric view of Linear Actuation mechanism

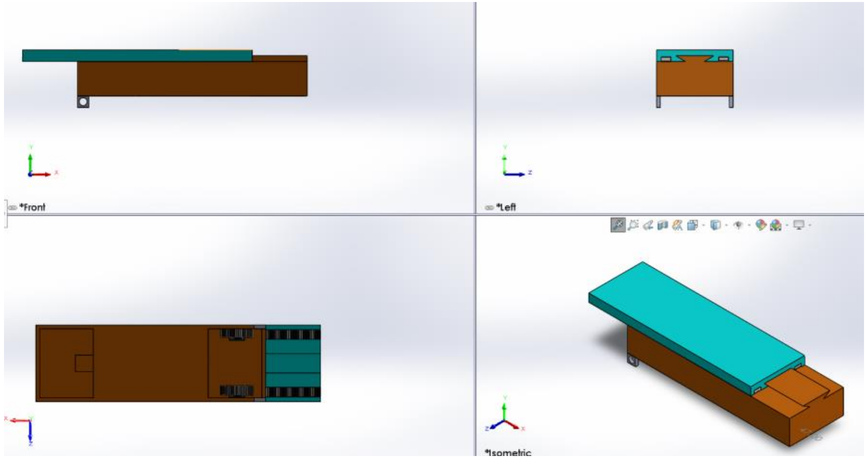


Figure9 : Orthogonal view of Rack and pinion mechanism

Worm Gear assembly

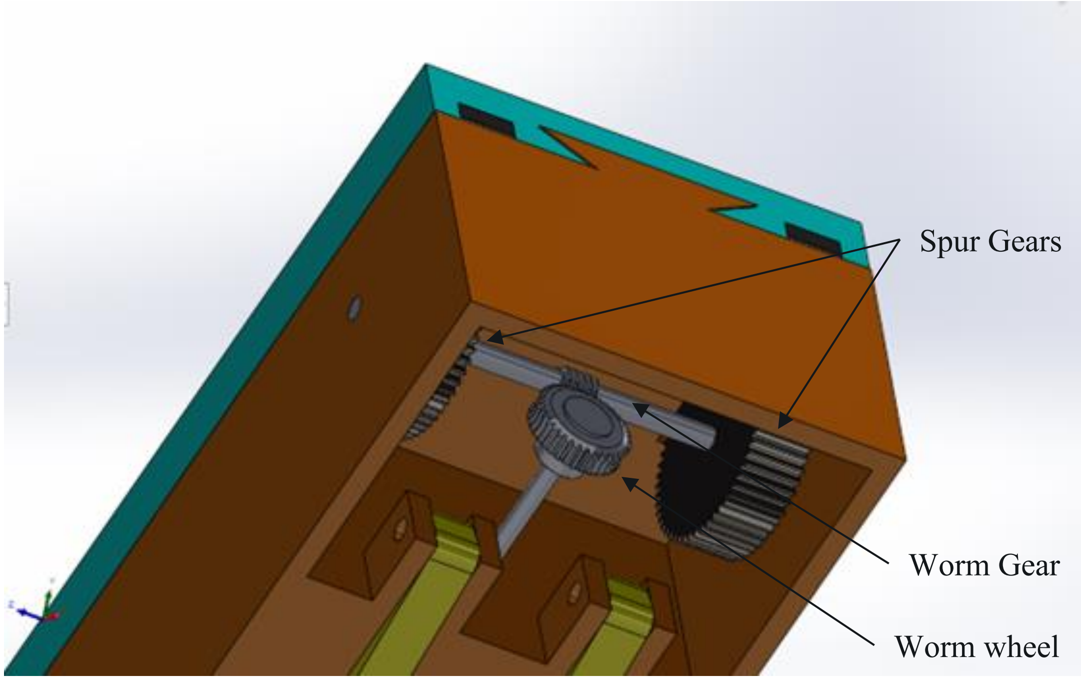


Figure10 : Isometric view of worm gears transmission assembly

Drawback of this system is:

- Worm gear can drive worm wheel but vice versa is not possible.
- This design failure can restrict the motion when there is a reflex action of the knee or disturbance in system.

Bevel Gear assembly

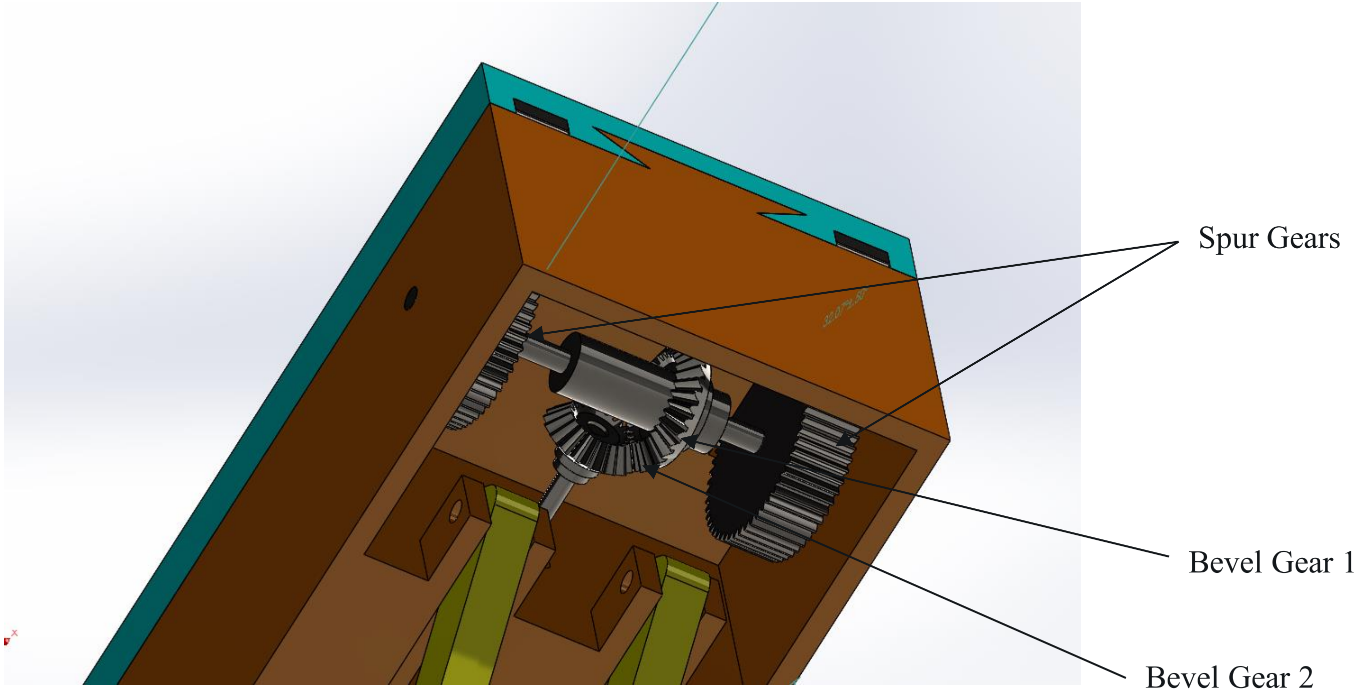


Figure10 : Isometric view of Bevel gears transmission assembly

Angular Actuation-

Lead Screw Mechanism with connecting rods

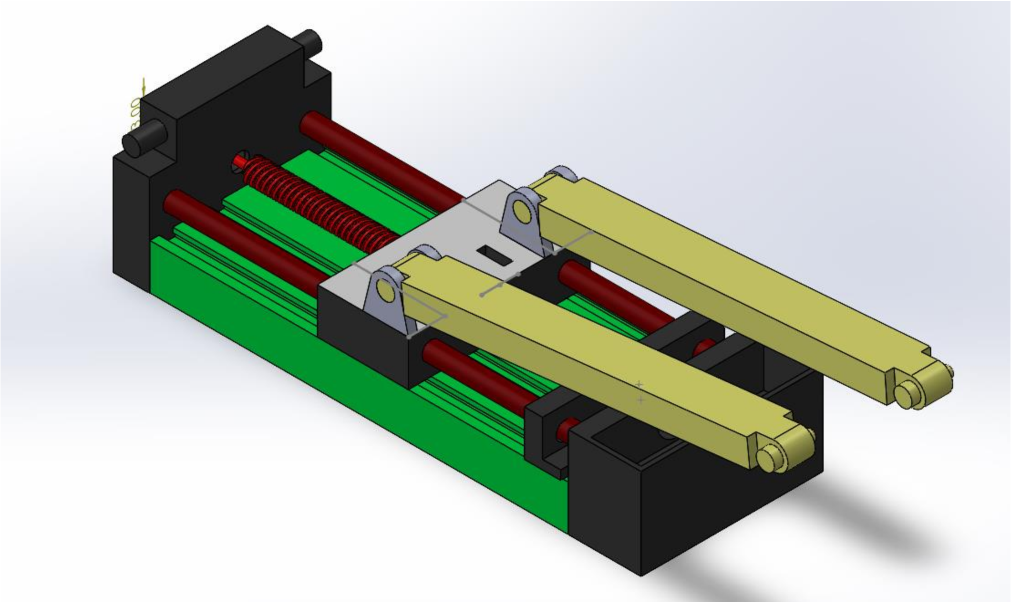


Figure11 : Isometric view of Angular Actuation mechanism

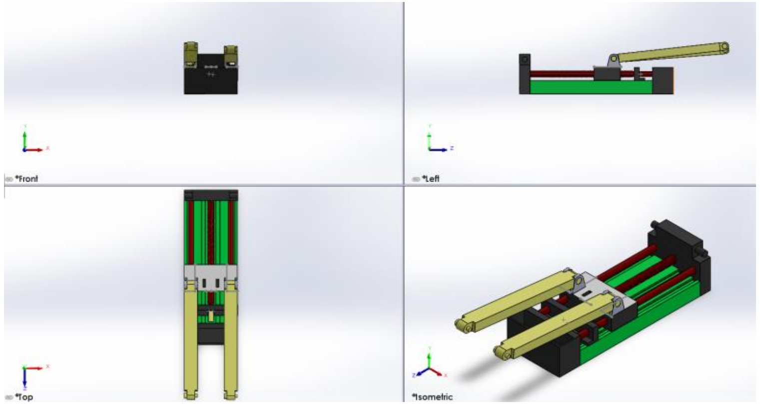


Figure12 : Orthogonal view of Lead screw mechanism with connecting rods

3rd modified design

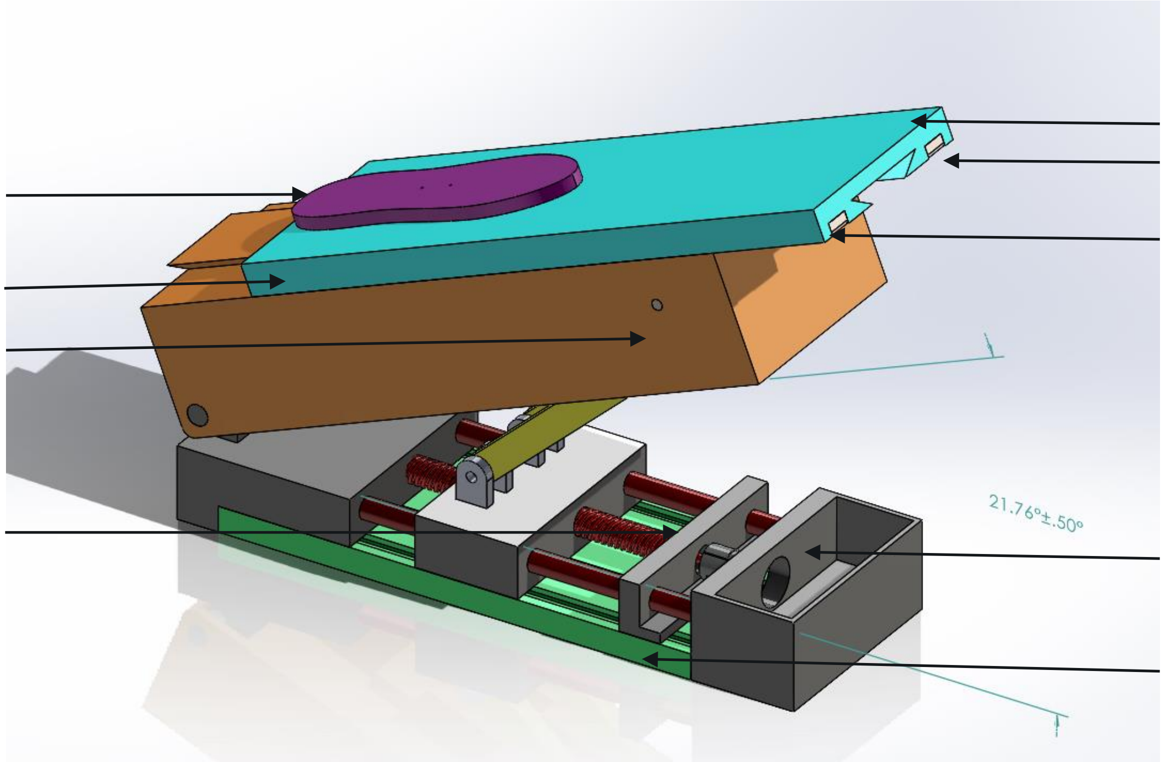


Figure13 : Isometric view of design 3

Orthographic projection of 3rd modified design

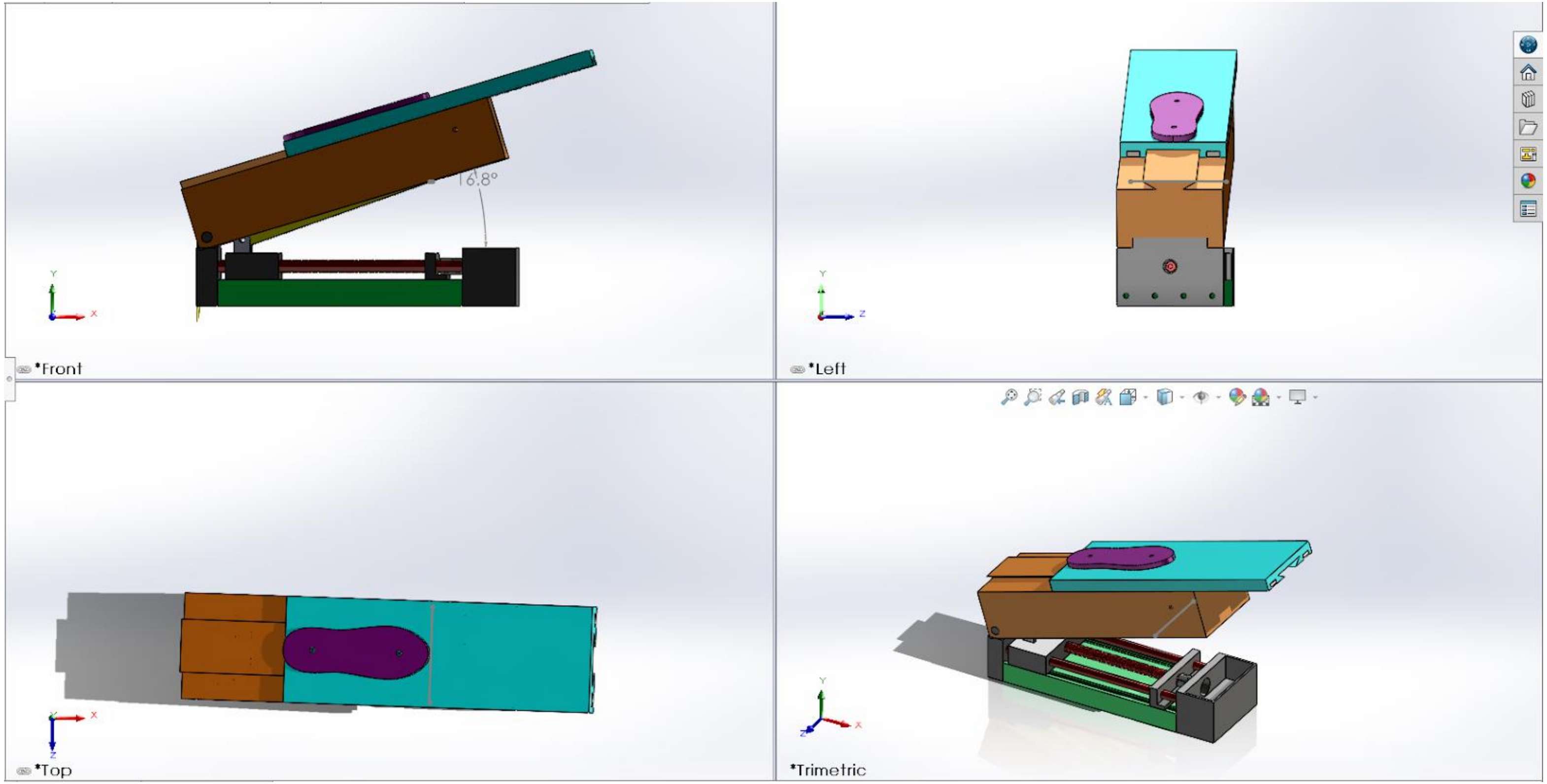


Figure14 : Orthographic projection of design 3

Structural analysis of 3rd Conceptual design

HOME POSITION

Total deformation

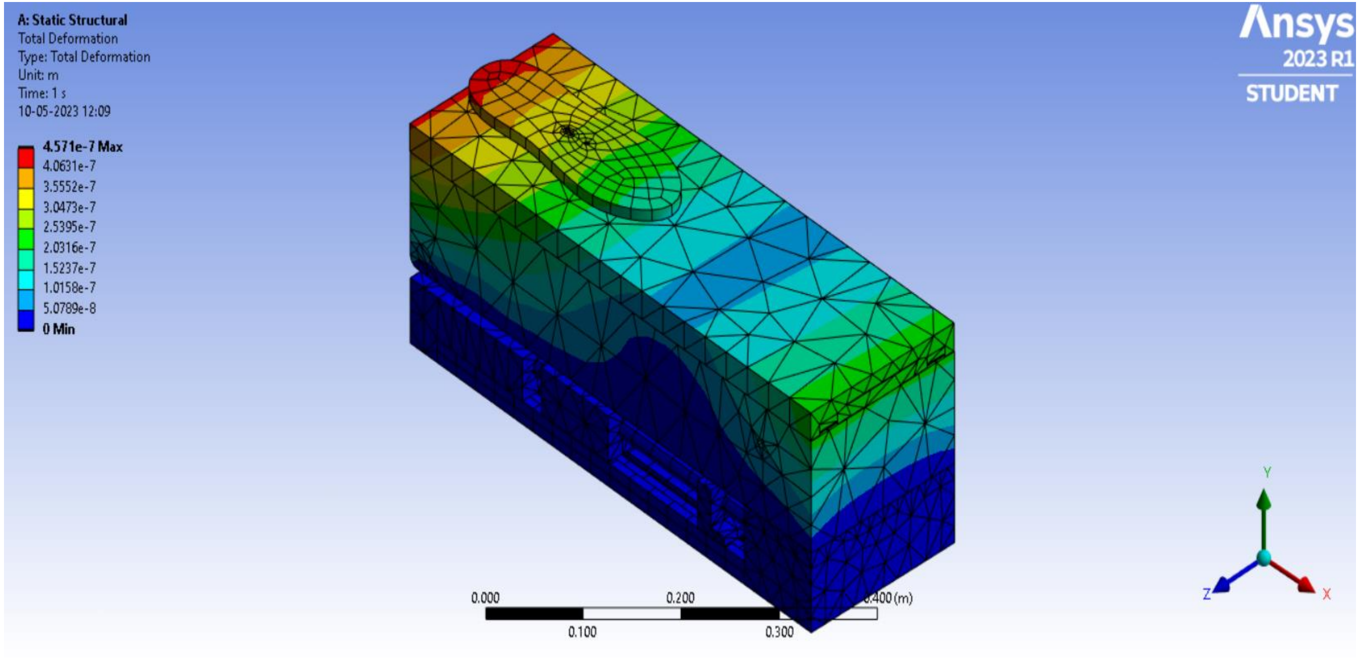


Figure15 : ANSYS interface of design 3

Structural analysis results on modified design

HOME POSITION

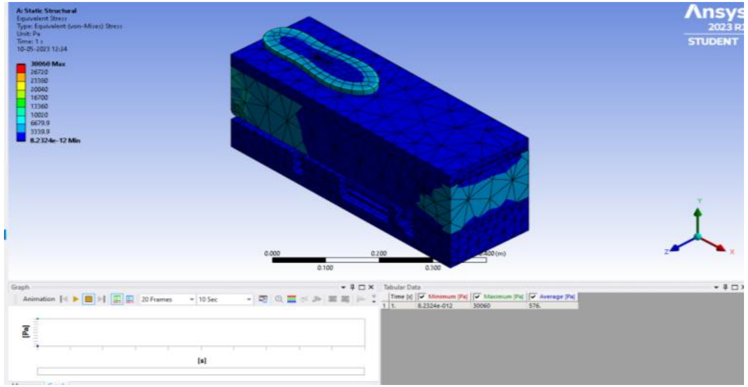
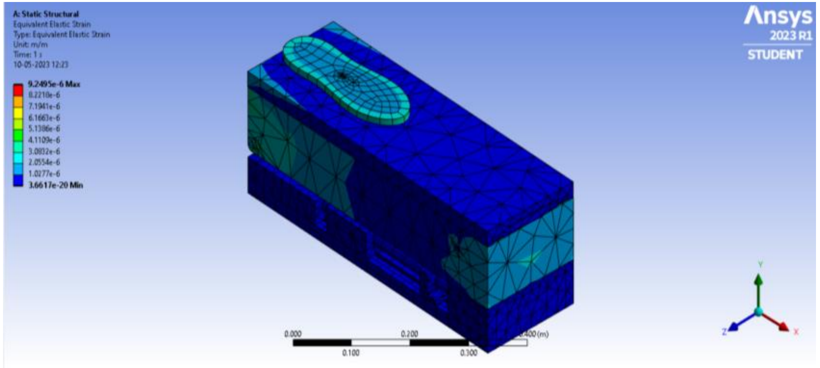


Figure16 : ANSYS interface of design 3

Equivalent stress



Equivalent Elastic strain

Figure17 : ANSYS interface of design 3

Structural analysis results on modified design

INTERMEDIATE POSITION

Total deformation

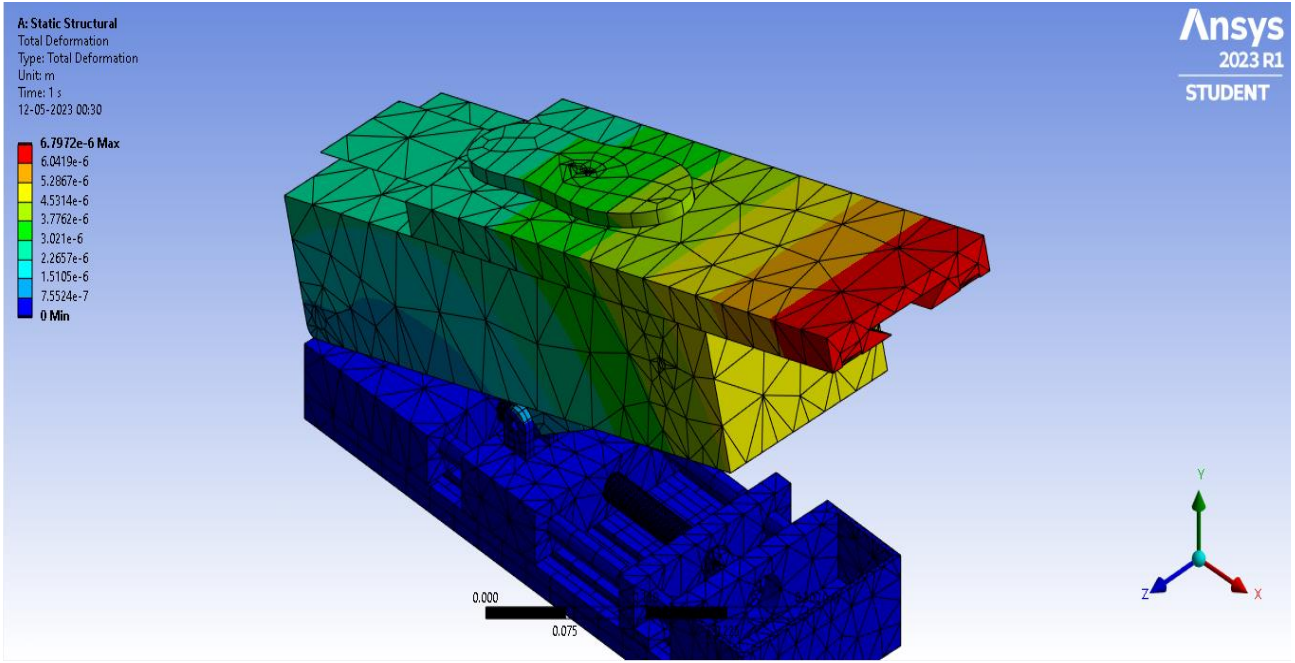


Figure18 : ANSYS interface of design 3

Structural analysis results on modified design

INTERMEDIATE POSITION

Equivalent stress

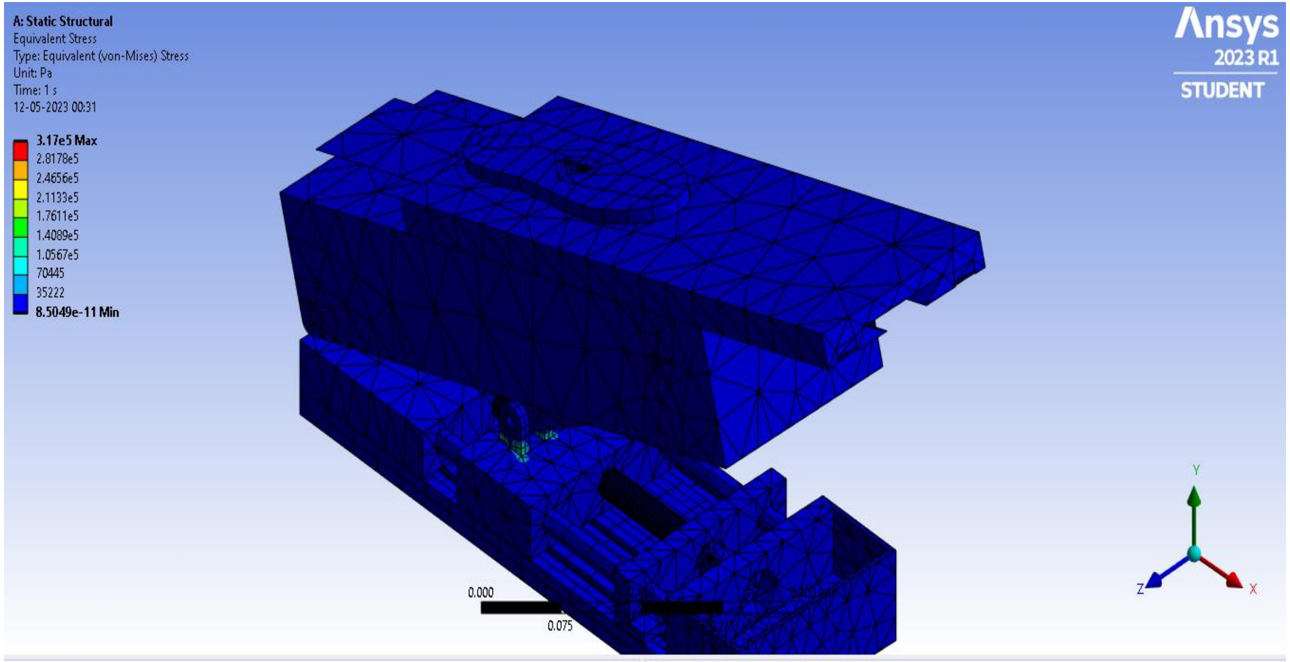


Figure19 : ANSYS interface of design 3

Structural analysis results on modified design

EXTENDED POSITION

Total deformation

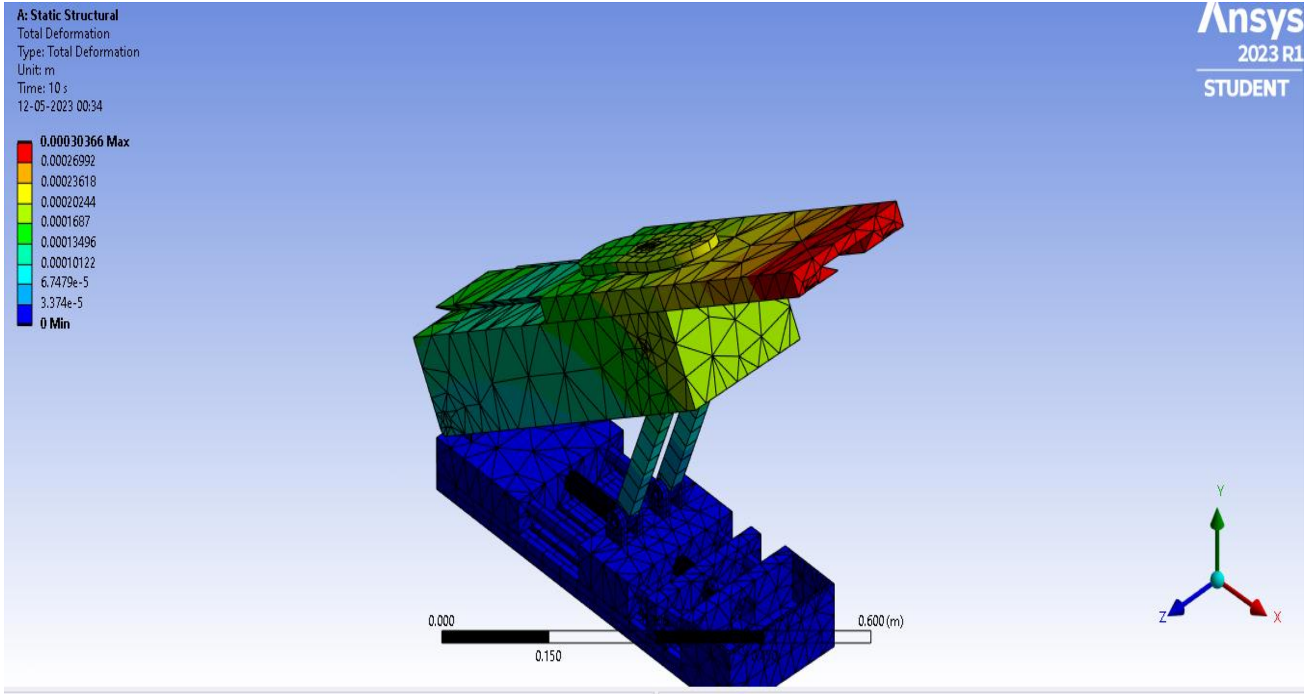


Figure 20 : ANSYS interface of design 3

Structural analysis results on modified design

EXTENDED POSITION

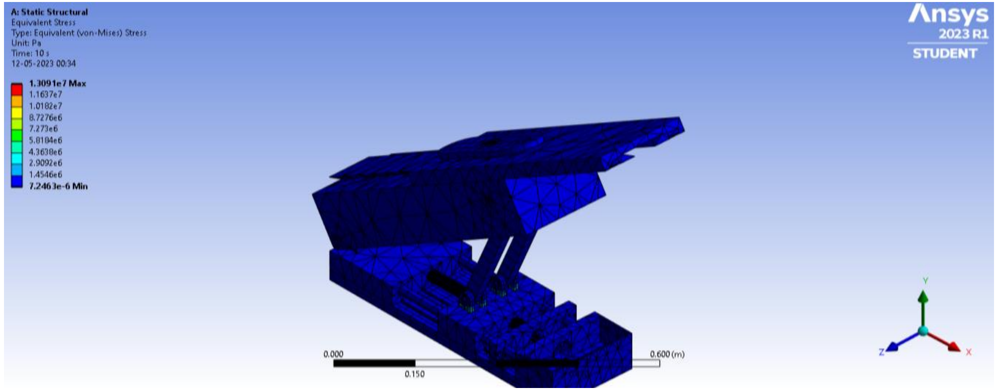


Figure 21 : ANSYS interface of design 3

Equivalent stress

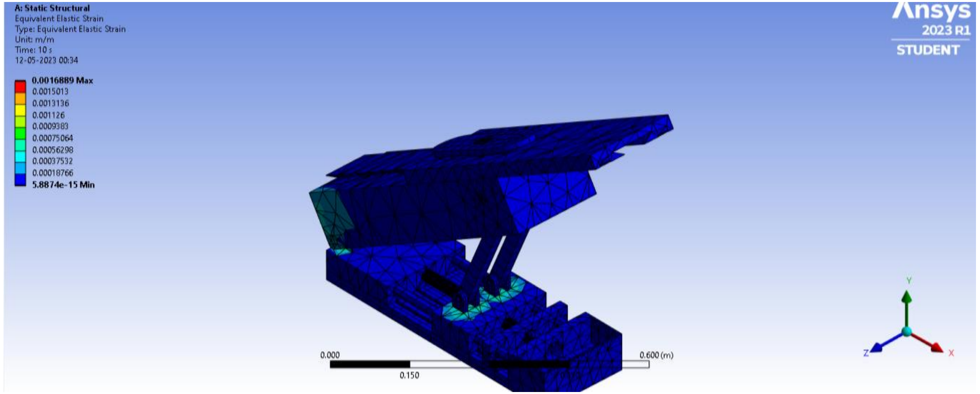


Figure 22 : ANSYS interface of design 3

Equivalent Elastic strain

MATLAB MODELLING

RACK MODELLING

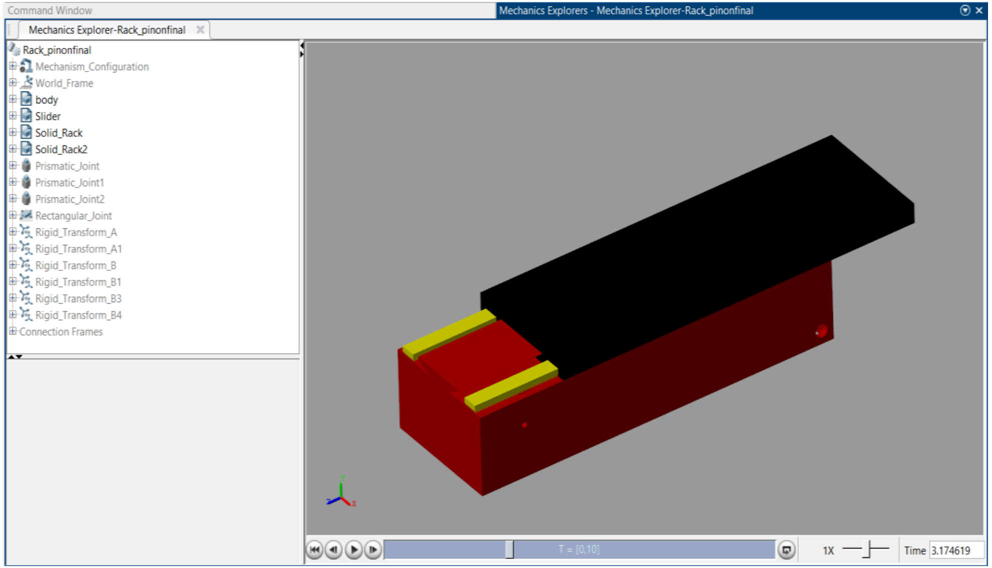


Figure 23 : Simscape multibody slider mechanism

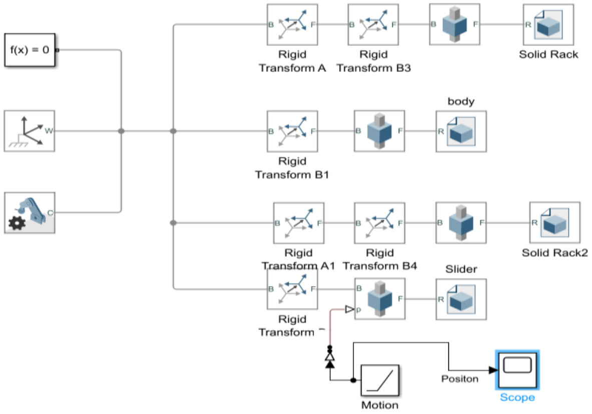


Figure 24 : Simulink slider mechanism

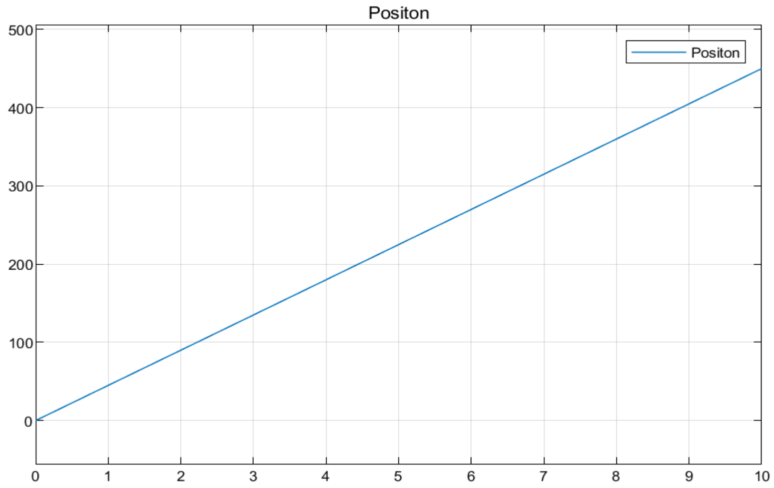


Figure 25: Position of slider w.r.t. time for slider mechanism

MATLAB MODELLING

RACK AND PINION MODELLING

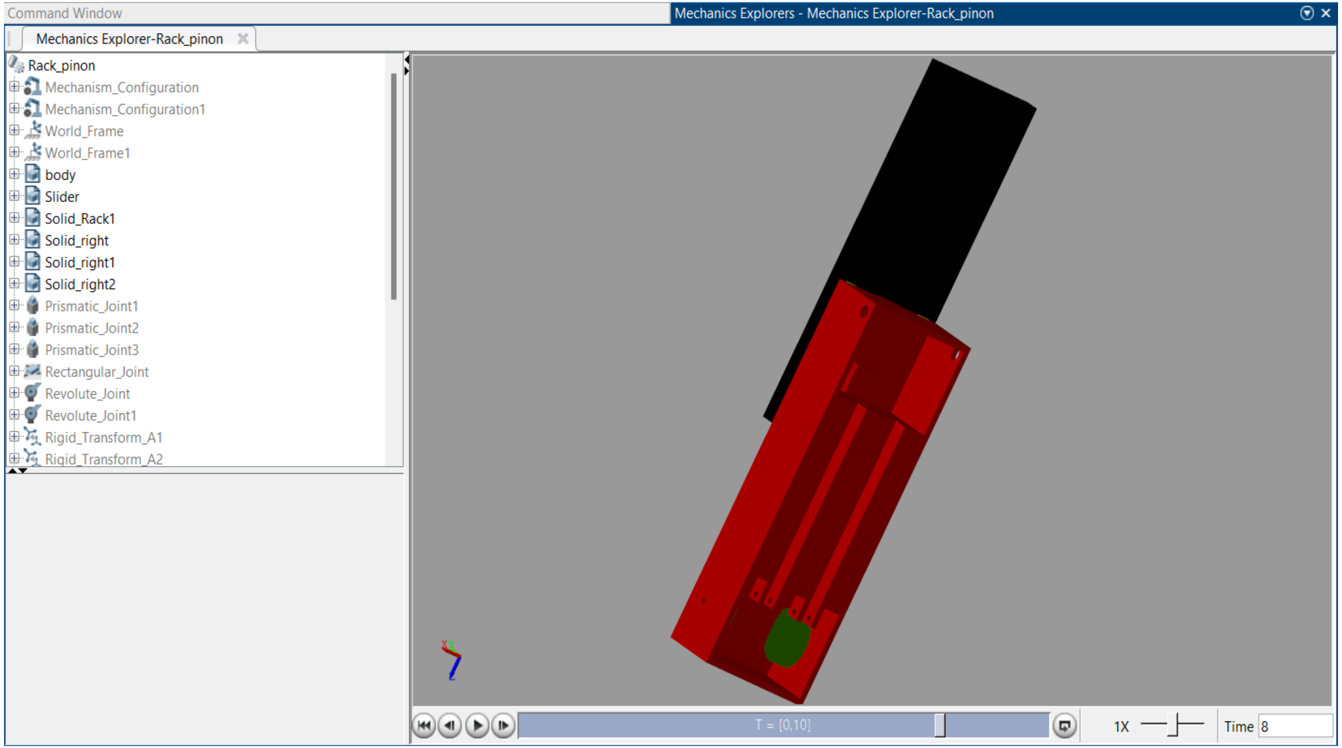


Figure 26 : Simscape multibody of rack and pinion mechanism

MATLAB MODELLING

RACK AND PINION MODELLING

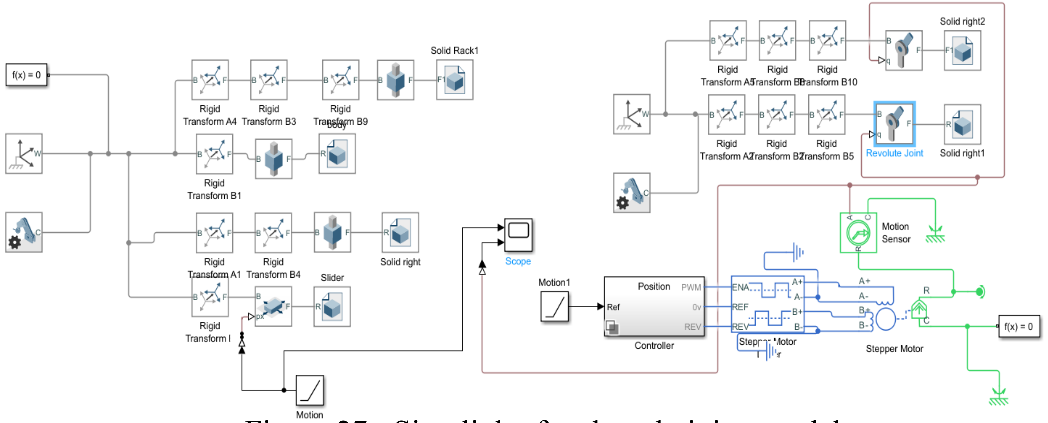


Figure 27 : Simulink of rack and pinion model

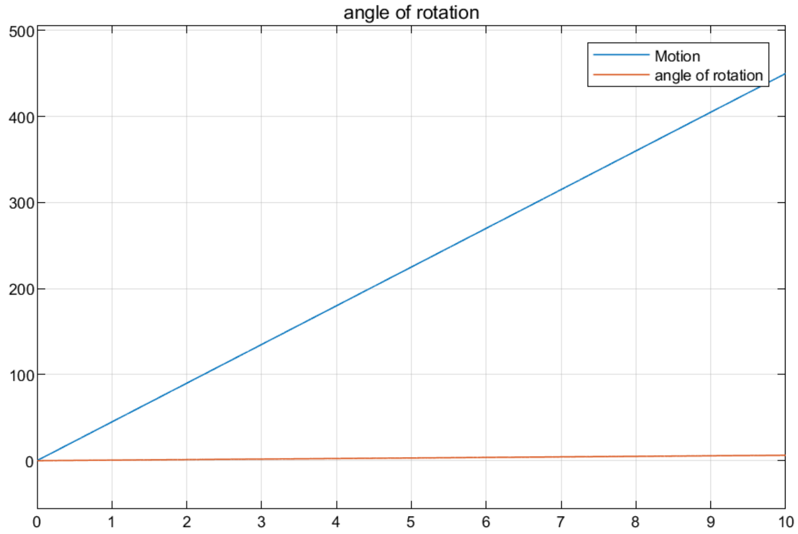


Figure 28: position and motion of stepper motor w.r.t. time for rack and pinion model

MATLAB MODELLING

LEAD SCREW MODELLING

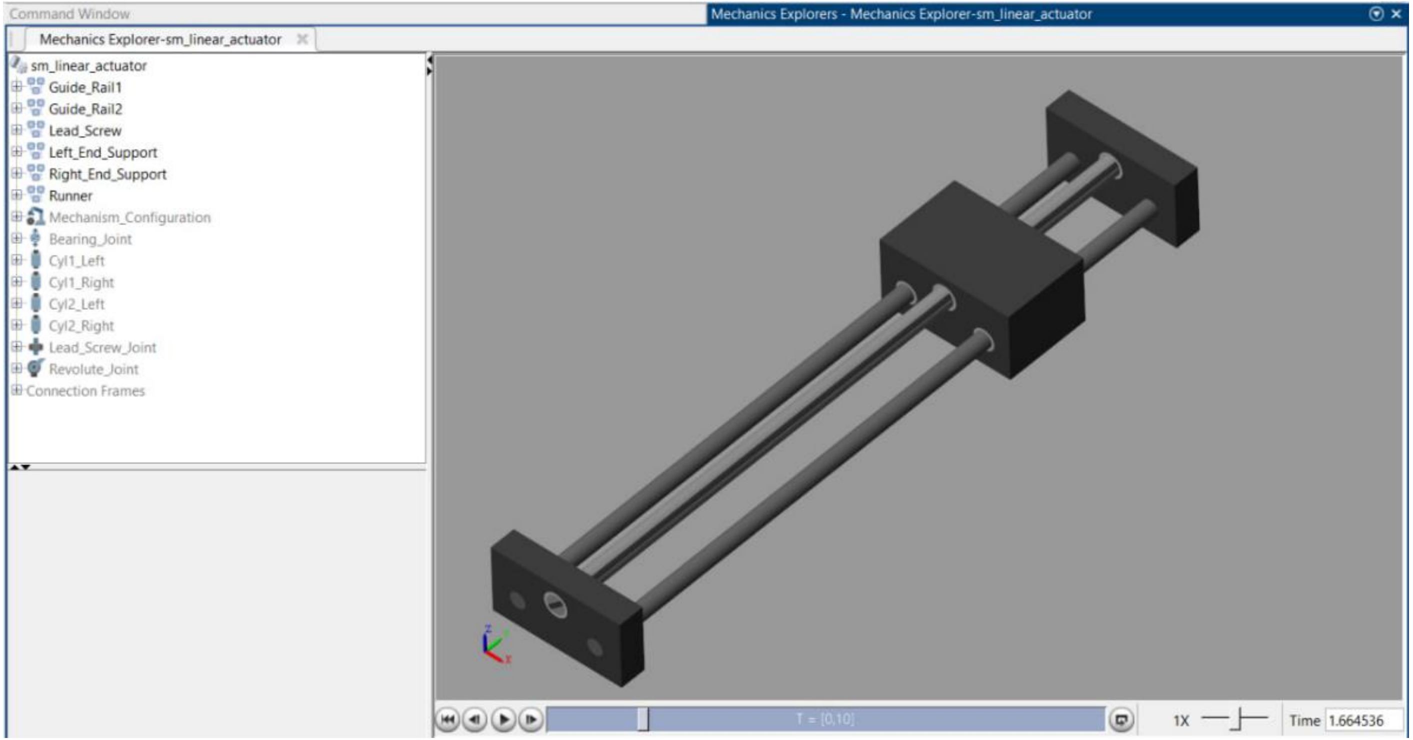


Figure 29 : Simscape multibody of lead screw model

MATLAB MODELLING

LEAD SCREW MODELLING

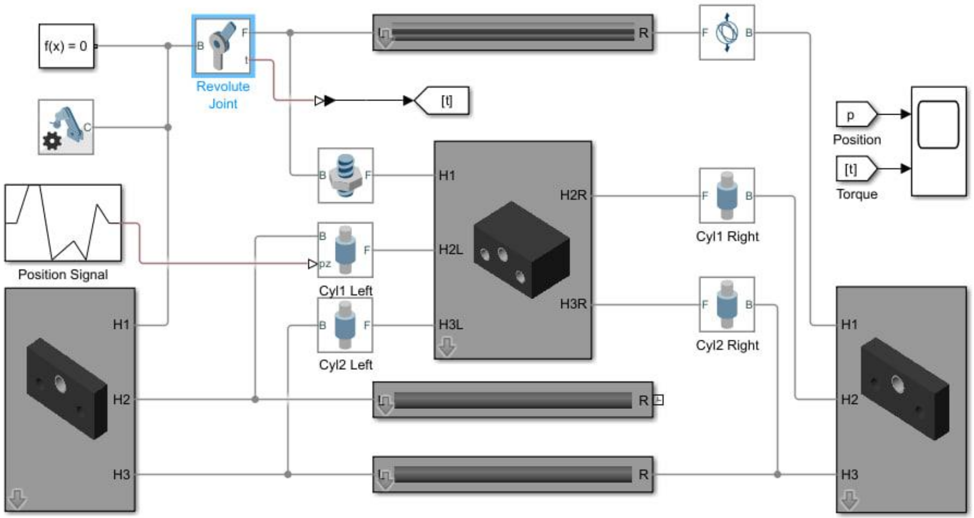


Figure 30 : Simulink of lead screw model

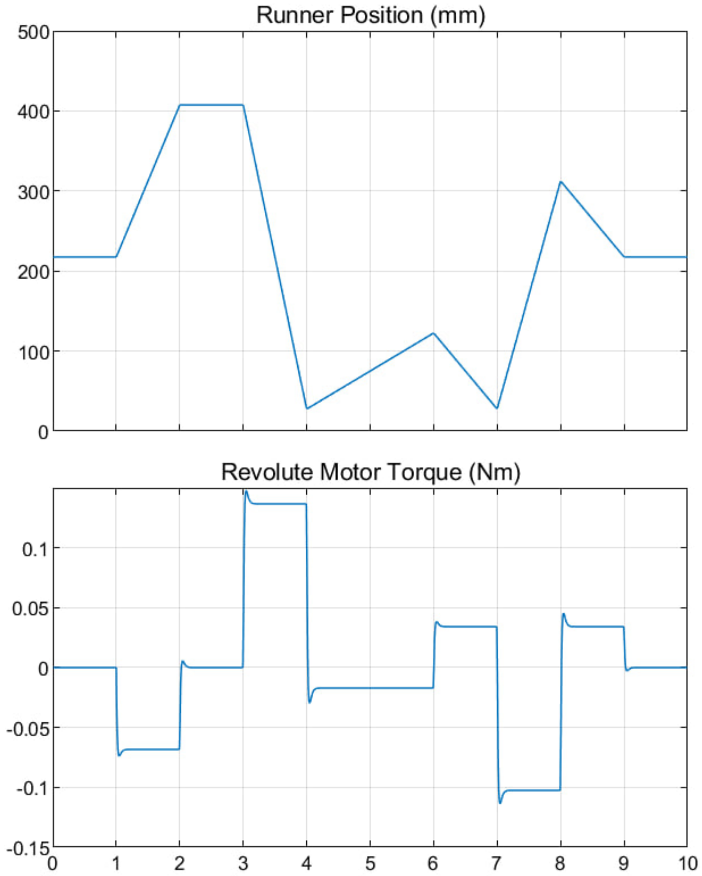


Figure 31: positon and torque of motor w.r.t. to time for lead screw mechanism

CALCULATIONS AND RESULTS

	Age group	Weight(kg)	Height(cm)	Foot length(cm)	Weight of lower leg(kg)
	0-20	52-100	153-172	16.5-19	8.14-14.70
	21-40	57-130	154-186	16-19	8.57-14.29
	41-60	65-78	152-173	16-19.5	7.64-12.14
Min-max	—	52-100	152-186	16-19.5	7.64-14.70

Table 2 : Collected Sample data of lower limb parameter

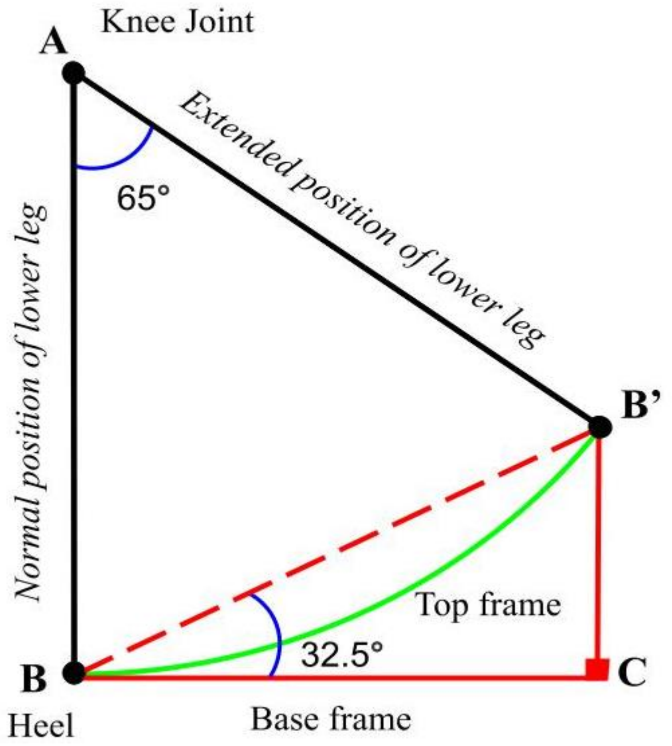


Figure 32: Line diagram of flexion movement

CALCULATIONS AND RESULTS

lower gait movement= 5 flexion/min
slider covers= 45cm
= 0.45/6
Speed of slider= 0.075m/s

HOME POSITION

Normal force applied on the foot plate = 45N

EXTREME POSITION

Normal force applied on the foot plate = 115 N
Frictional coefficient between two thermoplastic surfaces= 0.3

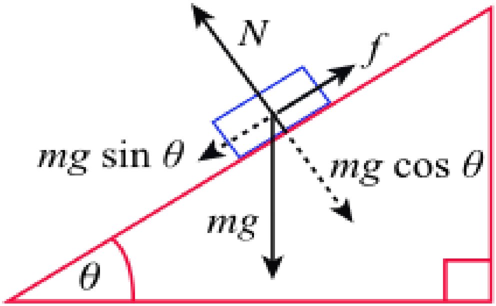


Figure 33: free body diagram on inclined plane

Mathematical modeling of Rack and Pinion

HOME POSITION

Torque of pinion $\approx$ 20 kg-cm

EXTENDED POSITION

Torque of pinion $\approx$ 50 kg-cm

Mathematical modeling of Lead Screw mechanism

Torque = 8.033 N-m (or 80.33 kg-cm)

Calculation- rack and pinion

Application		Horizontal movement	Vertical movement
	Unit	Application settings	
Total weight load	m	kg	kg
Speed	v	m/s	m/s
Time acceleration	ta	s	s
Gravity	g	$/s_2$	$/s_2$
Friction coefficient	μ	-	-
Pitch circle pinion	d	mm	mm
Other forces	F	N	N
Safety factor	S_B	-	-

Formulas					
		$\propto = V / t\propto$	(m/s^2)	$\propto = V / t\propto$	(m/s^2)
Tangential force	F_N	$F_N = M * g * \mu + M * a + F$	(N)	$F_N = M * g * \mu + M * a + F$	(N)
Torque	T_N	$T_N = (F_N * d) / 2000$	(Nm)	$T_N = (F_N * d) / 2000$	(Nm)
Design torque	T_{NV}	$T_{NV} = T_N * S_B$	(Nm)	$T_{NV} = T_N * S_B$	(Nm)
Max. speed pinion	N_V	$N_V = (V * 19100) / d$	(rpm)	$N_V = (V * 19100) / d$	(rpm)

Resultant speed: 25.5 RPM

HOME POSITION

Resultant torque: 20 kg-cm

EXTENDED POSITION

Resultant torque: 50 kg-cm

Module calculation of rack and pinion

To calculate Rack MOD (Module):

a) Measure distance of 10 (ten) Pitches, as shown below
b) Divide this number by 10 (ten)
c) Then divide the resulting number again by π (pi), this would give us the required Module.

E.g.

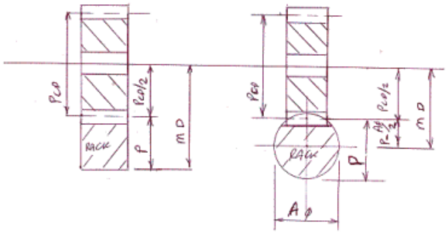
$$\text{Module} = \frac{\left(\dfrac{\text{Distance of 10 Pitches mm}}{10}\right)}{\pi}$$

Distance of 10 Pitches = 78.53 mm

Distance of one Pitch = 7.853 mm

$$\text{Required Mod} = \frac{7.853}{\pi} = 2.5$$

Rack Pinion Mounting Distance Sketch



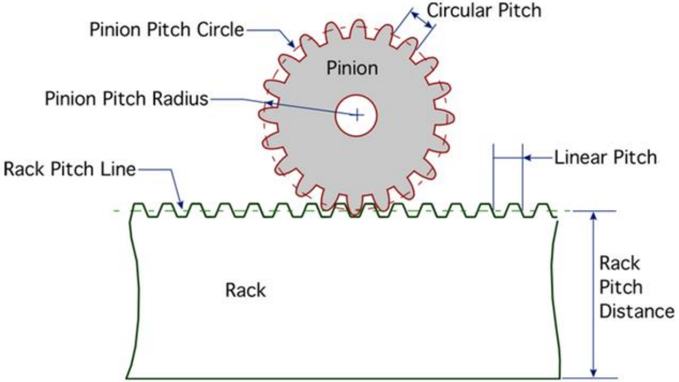
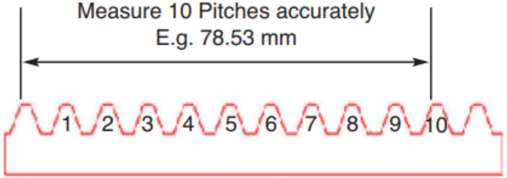


Figure34 : rack and pinion



Resultant module=3

Souce: https://cogmatic.com/rack-and-pinion-gear-sets/;https://www.hpcgears.com/pdf_c33/rack_pinion.pdf

FINAL DESIGN WITH DUMMY

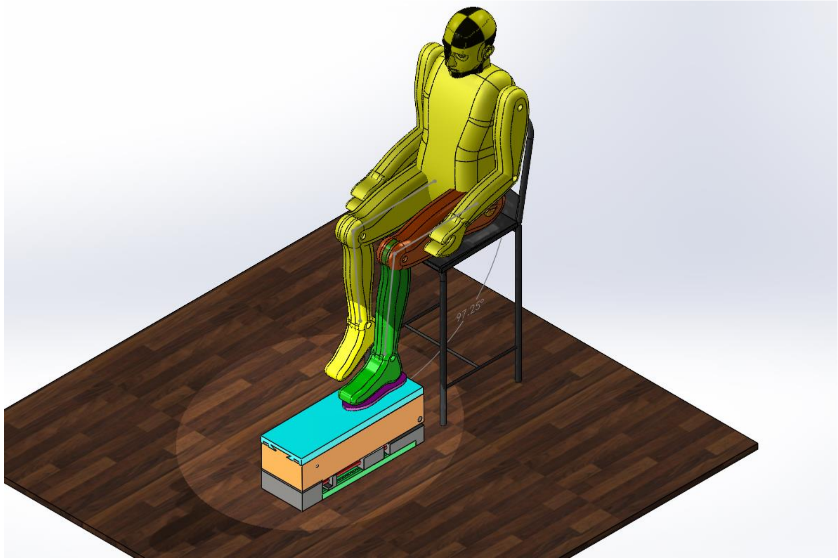


Figure 35: shows knee rehabilitation robot is in home position

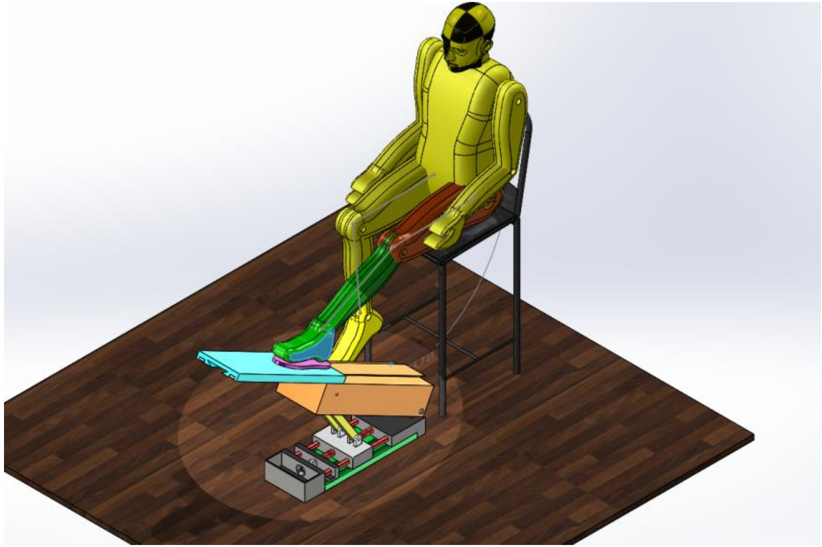
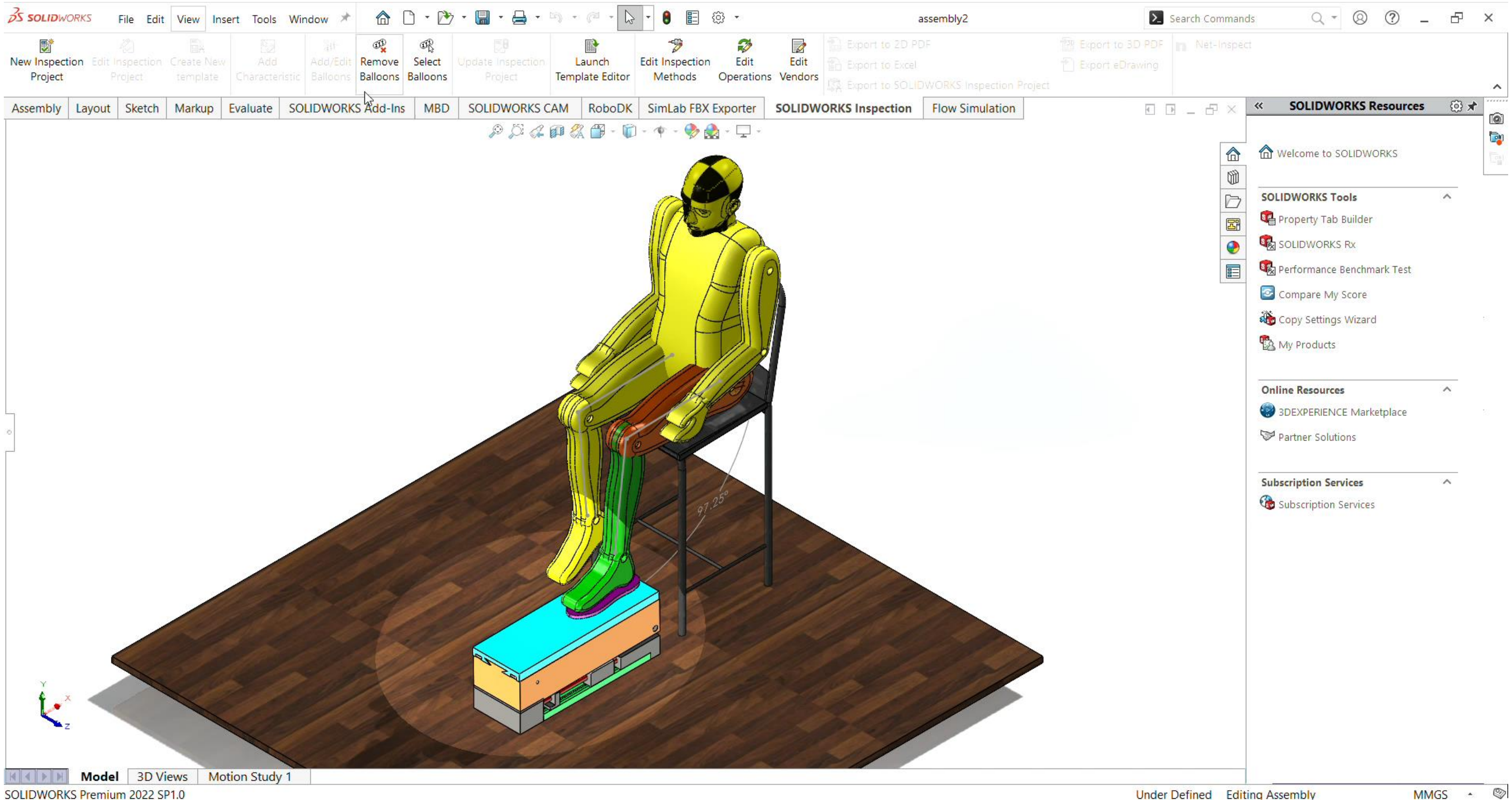
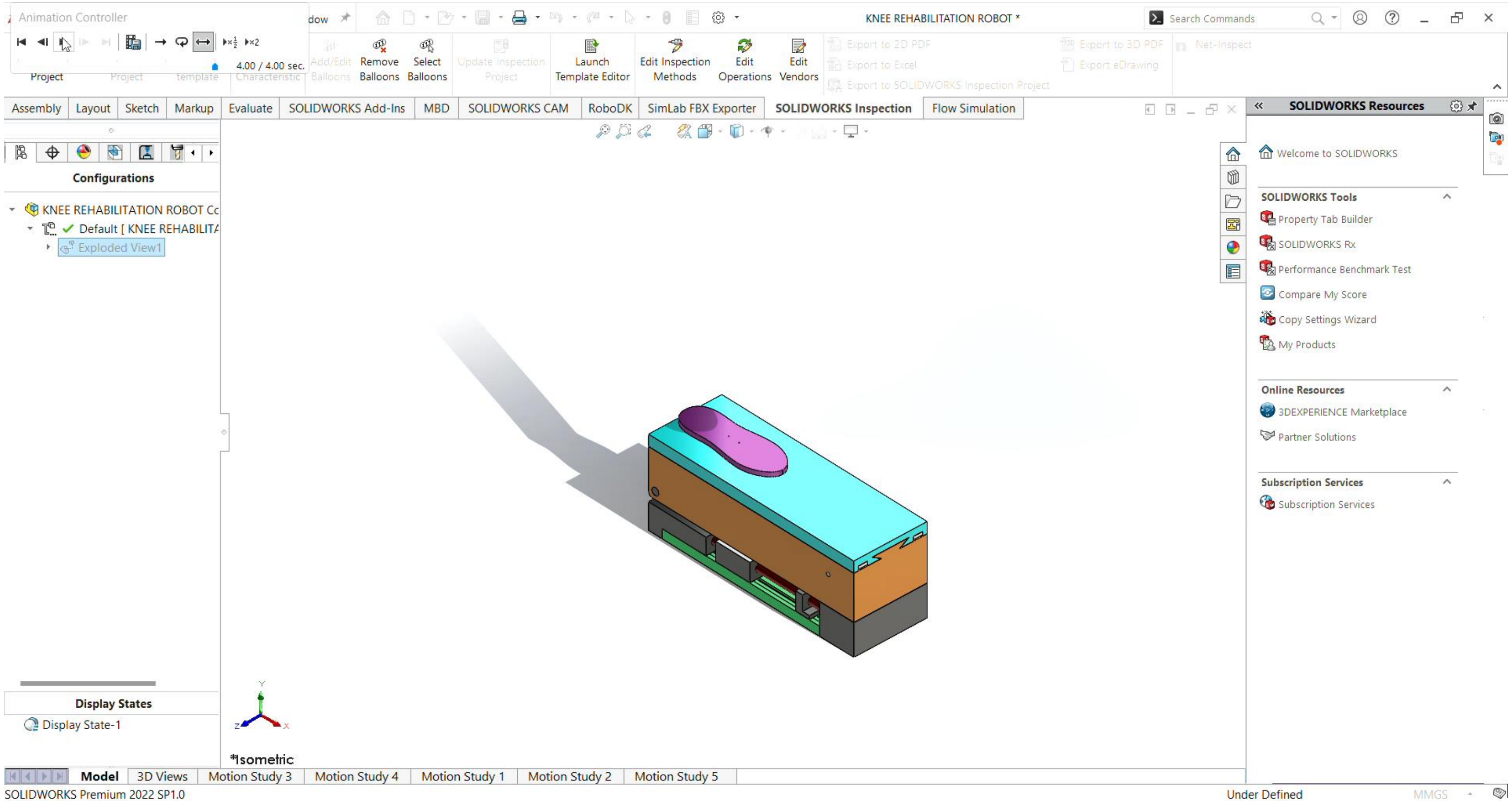


Figure 36: knee rehabilitation robot is in extended position

Rendering


SOLIDWORKS
File Edit View Insert Tools Window
assembly2
Search Commands
New Inspection Project
Remove Balloons
Select Balloons
Launch Template Editor
Edit Inspection Methods
Edit Operations
Edit Vendors
Assembly Layout Sketch Markup Evaluate SOLIDWORKS Add-Ins MBD SOLIDWORKS CAM RoboDK SimLab FBX Exporter SOLIDWORKS Inspection Flow Simulation
SOLIDWORKS Resources
Welcome to SOLIDWORKS
SOLIDWORKS Tools
Property Tab Builder
SOLIDWORKS Rx
Performance Benchmark Test
Compare My Score
Copy Settings Wizard
My Products
Online Resources
3DEXPERIENCE Marketplace
Partner Solutions
Subscription Services
Subscription Services
Model 3D Views Motion Study 1
SOLIDWORKS Premium 2022 SP1.0
Under Defined Editing Assembly MMGS

Rendering



Animation Controller
4.00 / 4.00 sec.
KNEE REHABILITATION ROBOT *
Search Commands
Remove Balloons
Select Balloons
Launch Template Editor
Edit Inspection Methods
Edit Operations
Edit Vendors
Assembly
Layout
Sketch
Markup
Evaluate
SOLIDWORKS Add-Ins
MBD
SOLIDWORKS CAM
RoboDK
SimLab FBX Exporter
SOLIDWORKS Inspection
Flow Simulation
SOLIDWORKS Resources
Configurations
KNEE REHABILITATION ROBOT Co
Default [KNEE REHABILITA
Exploded View1
Welcome to SOLIDWORKS
SOLIDWORKS Tools
Property Tab Builder
SOLIDWORKS Rx
Performance Benchmark Test
Compare My Score
Copy Settings Wizard
My Products
Online Resources
3DEXPERIENCE Marketplace
Partner Solutions
Subscription Services
Subscription Services
Display States
Display State-1
*Isometric
Model
3D Views
Motion Study 3
Motion Study 4
Motion Study 1
Motion Study 2
Motion Study 5
SOLIDWORKS Premium 2022 SP1.0
Under Defined
MMGS

3D PRINTED PROTOTYPES

1st PROTOTYPE

Procress - FDM
Material - PLA


(a)


(b)


(c)


(d)

Figure 37: prototype 1 3d printed

3D PRINTED PROTOTYPES

2st PROTOTYPE

- Process – FDM
- Material - PLA

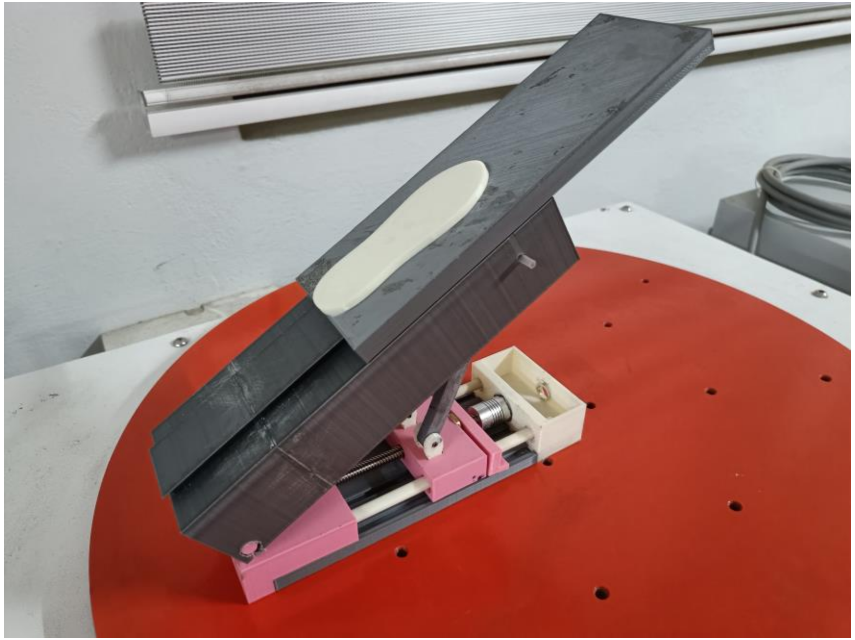
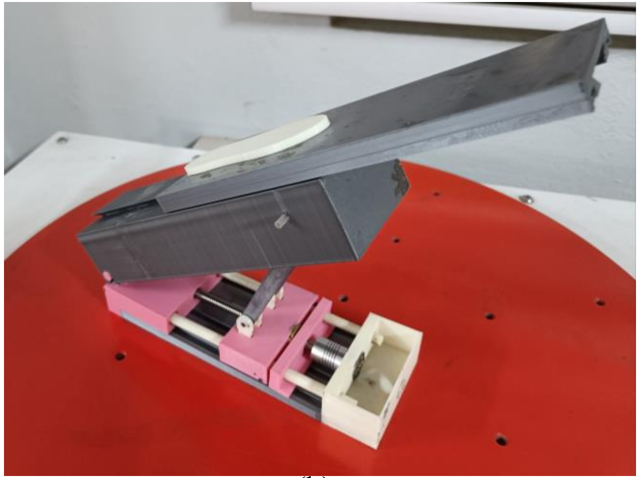


Figure 37: prototype 1 3d printed (a)



(b)

(c)

Literature Survey

Title	**Author**	**Inference**	**Methodology**
Development And Control Of A New Sitting-type Lower Limb Rehabilitation Robot, 2017	J.K. Mohan , S. Mohan , P. Deepasundar , R. Kiruba-shankar	•Use of prismatic joints for smooth limb motion •Use of parallel manipulators for lower limb actuation	• Kinematic and dynamic analyses PID control scheme
Knee rehabilitation using an intelligent robotic system, 2009	Akdoğan, E., Taçgın, E., & Adli, M. A.	•The robot manipulator works based on impedance control which is a control method for physiotherapy	The System has a servo motor and its driver as actuator, force/torque sensor and its controller for measurement of force data that come from therapist and patient and a data acquisition card for analog to digital–digital to analog conversion.

Title	Author	Inference	Methodology
Knee rehabilitation device with soft actuation: An approach to the motion control, 2018	Solaque, L., Romero, M., & Velasco	The feedforward strategy compensates loading disturbances while the feedback strategy acts in a defined threshold to maintain the stiffness of the system.	This paper tackles an approach to controlling a soft actuated rehabilitation device. It aims to control the link angular position of the knee rehabilitation device, to comply with defined repetitive motions
Musculoskeletal Model For Path Generation And Modification Of An Ankle Rehabilitation Robot, 2020	Prashant Jamwal, Sheng Quan Xie	•Path planning of ankle movement. •Quantitative analysis of link and joins	• Markovian jump linear system is used in design of force control

Title	Author	Inference	Methodology
Design and control of a lower limb rehabilitation robot considering undesirable torques of the patient's limb.	Almaghout, K., Tarvirdizadeh, B., Alipour, K., & Hadi, A.	The robot has a force controller unit with two controllers one for each leg. Using two controllers helps to receive data from two force sensors to operate quickly when controlling four motors	The robot uses the linear motion mechanisms in which the robot lifts and lowers the legs by kinematic interpretation, and the force sensor is attached to a device for pushing up the calf and thigh, measuring the pushing force, including the weight of the lower limb.
Rehabilitation of total knee arthroplasty by integrating conjoint isometric myodynamia and real-time rotation sensing system.	Luo, J., Li, Y., He, M., Wang, Z., Li, C., Liu, D., An, J., Xie, W., He, Y., Xiao, W., Li, Z., Wang, Z. L., & Tang, W.	Two sensing modules are mounted onto braces, 1)isometric myodynamia mea-surement (real-time force analysis) 2) active range of joint movement measurement (real-time angle analysis).	Use Active Rotation Sensing Module, Isometric Myodynamia Measurement Module and effectively get data online

Title	Author	Inference	Methodology
The use of robotics devices in knee rehabilitation: a critical review.	Wilmart, R., Garone, E., & Innocenti, B.	Advantages and different types of the robots used for rehabilitation.	By using Continuous Passive Motion (CPM) machines and Therapeutic Exercise Machines (TEM); • Gait-training Robots; • Exoskeletons.
An intelligent wearable robotic device for knee rehabilitation. ICTACT Journal on Soft Computing.	Prabakar, S., Porkumaran, K., Ramanan, S., Cheranmadevi, K., & Karthikeyan, R.	A new approach to ideology of assistive robots	Using an inertia motion sensor and relaying using Bluetooth Hc05

Problem Statement

Knee rehabilitation machines are sometimes used after knee replacement surgery to prevent scar tissue development, reduce stiffness, and restore normal function.

The main problem with those machines is their precision and affordability therefore we came up with a solution to develop a reliable solution for same.

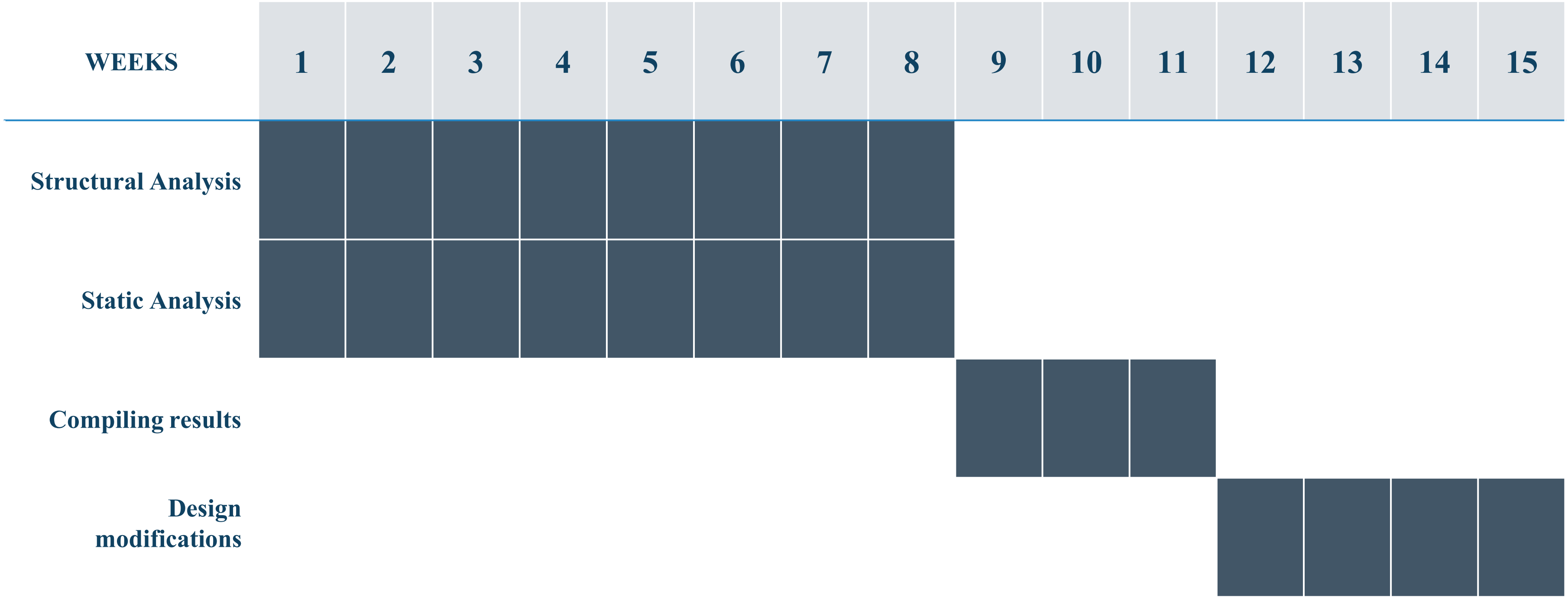
Obejctives & Scope

Our main objective is to design and develop a robotic system which will help in the process of knee rehabilitation.

Performance investigation and conceptual design optimization.

- Structural Analysis in ANSYS and mathematical modeling in MATLAB
- Design and simulation in SOLIDWORKS
- Designing control system for controlling 2 DOF of motion in MATLAB

GANTT CHART


WEEKS
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Structural Analysis
Static Analysis
Compiling results
Design modifications

DETAILS

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THANK YOU